

AI Transparency by Design: A Mixed-Methods Evaluation of Trust and Adoption in AI-Powered Learning Management Systems

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Abstract—Artificial intelligence (AI) is increasingly embedded in Learning Management Systems (LMSs) through content generation, adaptive recommendations, automated assessment, and tutoring. Prior studies demonstrate that these systems can improve course content quality, personalization, and learning support, but strong technical performance does not guarantee that users understand, trust, or intend to adopt AI-powered platforms. This study examines whether stronger AI-transparency user experience/user interface (UX/UI) features are associated with higher trust and adoption intention among university students and instructors. An explanatory sequential mixed-methods design with quantitative priority (QUAN→qual) is proposed. Approximately 10–15 students and 3–5 instructors would complete standardized tasks in Coursebox AI, LearnWorlds, MindSmith, TalentLMS, and Moodle with AI plugins. Each platform would be rated for four transparency features: AI labeling, explanations, source visibility, and uncertainty indicators. Participant trust and adoption intention would be measured with multi-item Likert-type scales, while the System Usability Scale (SUS) would provide a contextual measure of general usability. Quantitative findings would be analyzed using descriptive statistics, Friedman tests, follow-up paired comparisons, and exploratory linear mixed-effects models. Six to eight participants would then complete semi-structured interviews to explain high, low, or conflicting ratings. Expected findings are that stronger transparency will be associated with higher trust and adoption intention, while poor general usability may weaken adoption even when AI processes are relatively transparent.

Index Terms—artificial intelligence, learning management systems, AI transparency, UX/UI, trust, technology adoption, explainable AI, mixed methods

I. INTRODUCTION

Artificial intelligence (AI) is becoming increasingly integrated into higher education through personalized learning, automated content generation, intelligent tutoring, feedback, and adaptive learning systems. Learning Management Systems (LMSs) are beginning to incorporate these capabilities directly into platforms used by students and instructors. Research has shown that AI-based learning tools can improve personalization, engagement, and learning support [1]. However, as AI becomes more involved in educational content creation, concerns have also increased about whether users understand how AI-generated results are produced and whether they can trust those results.

Tian and Choo demonstrated the potential of generative AI for educational content generation by developing a system

that combines large language models, knowledge graphs, and learner modeling to produce personalized educational content [2]. Their study reported improvements in content quality, knowledge coverage, personalization, and content generation efficiency. These findings show that generative AI can make educational content creation more efficient and responsive to individual learner needs. However, the study focused primarily on the technical performance and educational effectiveness of the AI system rather than how students and instructors interpret or trust AI-generated content.

This distinction is important because strong technical performance does not necessarily lead to user trust or adoption. Research on AI adoption has shown that trust plays an important role in whether users are willing to accept and continue using AI technologies [3], [4]. Students may also recognize the usefulness of generative AI while still experiencing concerns related to usability, ethics, transparency, and uncertainty about AI-generated information [5]. In an LMS, users may encounter AI-generated recommendations, feedback, or instructional content without clearly knowing when AI was used, what information influenced the output, or how reliable the result may be.

User experience and user interface (UX/UI) design may help address these concerns. Generative AI interfaces can include transparency features that make AI processes more understandable to users [6]. Examples include clearly labeling AI-generated content, explaining how outputs or recommendations were produced, displaying sources, and communicating uncertainty or limitations. Explainable AI has also been integrated into educational platforms such as Moodle to provide more understandable and traceable AI-supported feedback [7]. These developments suggest that transparency should not only be considered a technical characteristic of an AI model but also a visible part of the interface.

Despite growing research on AI-powered educational systems, an important gap remains. Existing studies frequently focus on AI performance, personalization, learning outcomes, or general technology acceptance. Less research directly examines whether specific AI-transparency UX/UI features are associated with trust and willingness to adopt AI-powered LMS platforms. Understanding this relationship could help universities and LMS developers design AI systems that users

can evaluate more confidently and use more responsibly.

The purpose of this study is to examine whether stronger AI-transparency UX/UI features are associated with greater trust and adoption intention among university students and instructors. The study focuses on four transparency features: AI labeling, explanations, source visibility, and uncertainty indicators. Five AI-powered LMS platforms would be compared to determine how strongly these features are implemented and how users respond to them. General usability would also be measured to help determine whether adoption may be influenced by usability in addition to AI transparency.

The study is guided by the following research questions:

RQ1: Are stronger AI-transparency UX/UI features in AI-powered LMS platforms associated with higher trust and adoption-intention ratings among university students and instructors?

The qualitative phase addresses:

RQ2: How do students and instructors explain the ways AI labels, explanations, source visibility, and uncertainty indicators shape their trust and willingness to adopt AI-powered LMS platforms?

The following hypotheses guide the quantitative phase:

H₀: The strength of AI-transparency UX/UI features in AI-powered LMS platforms is not significantly associated with user trust or adoption intention.

H₁: Stronger AI-transparency UX/UI features in AI-powered LMS platforms are significantly associated with higher trust and adoption-intention ratings among university students and instructors.

The independent variable is the strength of AI-transparency UX/UI features, measured through AI labeling, explanations, source visibility, and uncertainty indicators. The dependent variables are user trust and adoption intention, measured through participant ratings. By focusing specifically on transparency rather than general interface design, this study extends previous research on the technical benefits of generative AI and examines how AI-powered LMSs can communicate their capabilities in ways that support informed trust and adoption.

II. LITERATURE REVIEW

A. AI-Powered Learning Systems and Personalization

Research on AI in education has rapidly expanded from isolated intelligent tutoring tools to systems integrated into broader digital learning environments. Luo *et al.* reviewed 63 peer-reviewed studies of AI-based learning tools in higher education and organized them by design characteristics, AI roles, and learning outcomes [1]. The review found extensive use of AI for assessment, personalized feedback, recommendations, intelligent tutoring, and multimodal learning. It also emphasized that AI tools can function as “black boxes” from the learner’s perspective and recommended human-centered, transparent, inclusive, and ethical design. This is directly relevant to the present study because it connects technical system design to the way students experience AI-supported learning.

Yang and Luo’s AI-based course teaching system represents the continuing movement toward integrated AI support within course environments [8]. The system-design focus is useful for showing how AI can be incorporated into teaching workflows rather than used only as an external chatbot. However, a system can be functionally sophisticated without necessarily communicating its AI processes clearly to end users. For the present research, this distinction supports the need to evaluate not only what an AI-enabled course platform can do, but also what students and instructors can understand about what it is doing.

Tian and Choo developed an educational content auto-generation algorithm that combines generative AI with knowledge graphs and learner modeling [2]. Their architecture uses a layered large-language-model structure, structured educational knowledge, adaptive reasoning, and output optimization. The authors reported improvements in content quality, knowledge point coverage, learning-style adaptation, course completion, and learning interest. The paper provides a technical foundation for extending research on AI-supported educational systems from technical performance to user trust and transparency. Its central contribution is evidence that advanced AI can improve educational content production and personalization. Its limitation for the present research question is that performance metrics do not reveal whether users can identify AI-generated material, understand the basis of an output, verify sources, or judge uncertainty.

Ezzaim *et al.* examined an AI-based adaptive learning plugin implemented in Moodle with 102 Moroccan high-school students [9]. The study compared adaptive learning with conventional methods and used satisfaction and summative evaluation measures. Results indicated improved engagement and academic performance, and a large majority of participants expressed interest in using the adaptive approach again. This study demonstrates that an AI enhancement can be integrated into a familiar LMS and can positively influence users. At the same time, its emphasis is adaptive effectiveness rather than the interface-level transparency of the AI process. It therefore supports the value of AI-powered LMS functionality while leaving the transparency question open.

Gadzhimusieva *et al.* developed SpectrumSphere, an LMS that integrates AI with other technologies to support personalized education for autistic students [10]. Using an extended Technology Acceptance Model (TAM3) with 30 special-education teachers, the study found that perceived usefulness, perceived ease of use, and computer self-efficacy were important predictors of behavioral intention. This work is especially relevant because it demonstrates that adoption is shaped by user perception and interface experience, not only by technical capability. However, its specialized population and platform purpose differ from a general university LMS, and transparency features such as source visibility and uncertainty communication were not the central independent variables.

Across these studies, a consistent pattern emerges: AI can improve personalization, automate educational processes, and support learning, but technical success does not fully answer

whether users trust the system. The proposed study therefore moves from the question “Can AI improve LMS functions?” to the question “How should an LMS communicate its use of AI so that users can judge and adopt it responsibly?”

B. Trust and Adoption of AI

Trust is especially important when people use AI because they may not always understand how the system reaches its results. Dang and Lee reviewed 562 studies on trust in AI and found that factors such as system capability, user characteristics, explainability, and transparency can influence how users trust AI [4]. They also found that trust can affect attitudes, perceived usefulness, acceptance, and willingness to use AI systems. Their review suggests that transparency and explainability may play an important role in building trust, while trust itself may influence whether users are willing to adopt AI powered systems.

Marmolejo-Ramos *et al.* studied perceptions and adoption of AI among 1,906 university students across 13 countries [3]. They found that familiarity with AI, perceived effectiveness, and AI-related anxiety were important factors in how students viewed and adopted AI technologies. This study is especially relevant because it shows that trust is closely connected to students’ willingness to use AI in higher education, which supports the focus on trust and adoption in the current study. However, their research examined the general perceptions of AI and large language models rather than how specific interface features shape those perceptions. The current study builds on this work by focusing more specifically on whether visible AI transparency features, such as AI labels, explanations, source visibility, and uncertainty indicators, are associated with greater trust and adoption in AI-powered LMS platforms.

This connection between trust and willingness to use AI is also reflected in the work of Colbran *et al.*, who used chatbot analytics and a mixed-methods survey of 121 students to study generative AI chatbots in higher education [5]. These findings show that students can see value in an AI tool while still feeling uncertain about whether they should fully trust or rely on it. The study is also relevant methodologically because it shows how quantitative results can identify general patterns in student responses, while qualitative feedback can help explain why those patterns occur. The current study uses a similar mixed methods approach but focuses specifically on how AI transparency UX/UI features may shape trust and adoption in LMS platforms.

Together, these studies show that trust and adoption are closely related, but they are not the same. A user may trust the information provided by an AI system but still choose not to use the platform if the interface is difficult or frustrating. In the same way, a user may be willing to use a convenient AI tool even while having concerns about how reliable or transparent it is. For this reason, the current study measures trust and adoption intentions separately. It also uses SUS as a measure of general usability so that differences in adoption can be considered in relation to both AI transparency and the overall user experience.

C. Generative-AI UX/UI and Explainable AI in LMSs

Generative AI creates interface-design challenges that differ from conventional deterministic software. A traditional interface generally maps a user action to a predictable system response. Generative AI may instead return probabilistic, incomplete, or variable output. Users therefore need interface cues that communicate what the system generated, why it produced a result, what evidence supports the result, and where uncertainty remains.

Kim *et al.* developed a taxonomy of UI/UX for generative AI and examined trends and challenges across generative applications [6]. Their work is relevant because it places human-computer interaction at the center of generative-AI design. Instead of treating the model as the entire product, the study emphasizes that the way users prompt, interpret, evaluate, and revise AI output is shaped by interface design. For an LMS, this suggests that transparency cannot be left only in documentation or technical model cards. It must be presented at the point where students and instructors encounter AI-generated material.

Villegas-Ch *et al.* provide a concrete example of explainability embedded in an LMS [7]. Their Moodle-based educational assistant used BERT and CodeBERT for semantic analysis and LIME and SHAP for explainable feedback. The system was validated with 90 students and was designed to connect semantic assessment, explanation, and adaptive tutoring. The study demonstrates the technical feasibility of traceable AI-supported feedback inside a real LMS. However, its evaluation centers mainly on semantic accuracy, explanation coverage, tutoring appropriateness, and system performance rather than a comparative test of whether visible transparency design is associated with users’ trust and adoption intention.

This distinction is central to the proposed study. Explainable AI can exist technically without being communicated effectively through UX/UI. A back-end model may be explainable to developers, while a student sees only a polished recommendation with no source, confidence indicator, or explanation. The present study therefore evaluates user-visible transparency rather than only the existence of an explainability algorithm.

TABLE I
SUMMARY OF RELATED STUDIES, PART I

Study	Objective	Methods/Design	Framework	AI/ML	Key Findings	Limitations/Gap	Relevance
Luo <i>et al.</i> (2025) [1]	Synthesize design and outcomes of AI learning tools in higher education.	Systematic review of 63 peer-reviewed studies from 2014–2024.	Learning-outcome and human-centered design perspective.	Multiple techniques across reviewed systems.	AI commonly supports assessment, feedback, recommendations, tutoring, and multimodal learning; outcomes are generally positive but vary by skill.	Broad review; does not experimentally isolate specific LMS transparency features.	Establishes need for transparent, human-centered AI design in higher education.
Yang and Luo (2025) [8]	Design an AI-based course teaching system.	System-design/development study; human participant sample not central to the cited study.	Intelligent course teaching and system design perspective.	AI-supported course functions.	Demonstrates integration of AI into course teaching workflows.	Limited evidence about user adoption, or transparency UX/UI.	Shows the broader movement toward AI-integrated course platforms.
Tian and Choo (2025) [2]	Improve automatic educational-content generation and personalization.	Experimental system evaluation using large educational datasets and learner records; statistical comparison and cross-validation.	Cognitive load theory, Zone of Proximal Development, constructivist learning.	LLMs, Transformers, knowledge graphs, BERT-based extraction, Bayesian knowledge tracing, adaptive reasoning.	Reported improvements in content quality, knowledge coverage, personalization, efficiency, completion, and interest.	Focuses on technical/learning performance rather than user-facing AI transparency, trust, or adoption.	Provides a direct foundation for extending technical performance research to user trust and AI transparency.
Ezzaim <i>et al.</i> (2024) [9]	Compare AI-based adaptive Moodle learning with conventional learning.	Evaluation with 102 Moroccan high-school students; satisfaction survey and summative assessment.	Adaptive-learning perspective.	AI-based adaptive Moodle plugin.	Positive effects on engagement, motivation, performance, and willingness to reuse the approach.	High-school setting; transparency features not isolated.	Demonstrates benefits of adding AI functionality directly to an LMS.
Gadzhimusieva <i>et al.</i> (2026) [10]	Develop and evaluate adoption of a personalized LMS for autistic students.	Prototype/pilot evaluation with 30 special-education teachers.	Extended Technology Acceptance Model (TAM3).	AI plus VR, AR, robotics, AAC, and personalization.	Usefulness, ease of use, and computer self-efficacy strongly related to behavioral intention.	Specialized context and small teacher sample; transparency is not the main manipulated construct.	Shows that adoption depends on human factors and interface experience, not technical capability alone.

TABLE II
SUMMARY OF RELATED STUDIES, PART II

Study	Objective	Methods/Design	Framework	AI/ML	Key Findings	Limitations/Gap	Relevance
Marmolejo-Ramos <i>et al.</i> (2026) [3]	Examine university students' perceptions and adoption of AI/LLMs across countries.	Cross-national psychometric study; 1,906 students from 13 countries.	Technology perception/adoption and psychometric perspective.	LLM-assisted item generation/validation; ChatGPT, Claude, and Bard/Gemini used in instrument development.	Trust, algorithm familiarity, perceived effectiveness, and lower anxiety support more positive AI perceptions and adoption.	Convenience sampling and broad AI focus; does not compare LMS transparency designs.	Establishes trust as a central adoption factor in higher education.
Dang and Li (published online 2025; 2026 volume) [4]	Synthesize antecedents, consequences, and cultural dynamics of trust in AI.	Systematic review of 562 empirical studies using PRISMA-informed procedures.	Multidimensional trust and cultural framework.	Multiple AI system types across studies.	Explainability and transparency are important antecedents; behavioral intention and acceptance are major consequences.	Very broad domains and heterogeneous methods; not specific to university LMS interfaces.	Provides conceptual support linking transparency, trust, and adoption.
Colbran <i>et al.</i> (2026) [5]	Understand student perspectives on generative-AI chatbots in higher education.	Human-centered mixed-methods study using chatbot analytics and survey data ($n = 121$).	Human-centered perspective.	Generative-AI chatbots.	Students saw educational value but reported UX barriers, ethical concerns, and policy ambiguity.	Chatbots rather than full LMS platforms; cross-sectional self-report limits causal inference.	Supports mixed methods and the need for clear AI communication and usable interfaces.
Kim <i>et al.</i> (2024) [6]	Organize UI/UX patterns, trends, and challenges for generative AI.	Taxonomy/review and HCI-oriented analysis of generative-AI interfaces.	Human-computer interaction and UI/UX taxonomy.	Generative AI across text, image, audio, and multimodal systems.	Shows that AI interaction design must support user understanding, control, and evaluation of generated output.	Broad generative-AI scope; does not test trust/adoption in university LMSs.	Provides design logic for evaluating user-visible transparency patterns.
Villegas-Ch <i>et al.</i> (2025) [7]	Build an explainable educational assistant integrated in Moodle.	System development and controlled validation with 90 students and real academic submissions.	Explainable and adaptive-tutoring architecture.	AI BERT, LIME, CodeBERT, SHAP, adaptive tutoring index.	Demonstrates technically feasible semantic assessment, traceable explanations, and adaptive tutoring in Moodle.	Evaluation emphasizes technical performance and explanation coverage rather than comparative user trust/adoption.	Direct evidence that explainability can be embedded in LMS workflows.

D. Synthesis of the Literature and Remaining Gap

The literature supports four conclusions. First, AI-powered educational systems can improve personalization, feedback, content generation, and learning support [1], [2], [9]. Second, user adoption depends partly on perceived usefulness, ease of use, trust, familiarity, and other human factors [3], [4], [10]. Third, user-experience problems and uncertainty about AI can reduce confidence even when users recognize potential educational value [5]. Fourth, generative-AI UX/UI and XAI research show that explanation and transparency can be designed into interfaces and LMS functions [6], [7].

What remains unclear is whether AI-powered LMS platforms that provide stronger combinations of user-visible transparency features receive higher trust and adoption-intention ratings from university students and instructors. Existing studies typically examine one platform, one AI function, general AI attitudes, or technical explainability. Few directly compare several AI-enabled LMS platforms using a consistent transparency rubric and then ask users to explain why particular interface features affected their trust.

The proposed study addresses this gap in three ways. It operationalizes AI transparency into four observable UX/UI components; it compares five AI-powered LMS platforms using the same rating approach and standardized participant tasks; and it uses explanatory sequential mixed methods so that qualitative interviews are selected to explain quantitative patterns. The design also separates AI transparency from overall usability, allowing the study to examine whether a transparent system can still have low adoption because of a poor general user experience.

III. METHODOLOGY

A. Research Design

The proposed study would use an explanatory sequential mixed-methods design with quantitative priority, represented as QUAN→qual. In this design, the quantitative phase would first identify whether stronger AI-transparency UX/UI features are associated with higher trust and adoption-intention ratings. The qualitative phase would then follow the quantitative analysis and would be designed specifically to explain the patterns, exceptions, and contradictions observed in the first phase. This structure follows the logic of explanatory sequential mixed-methods research described by Creswell and Creswell and by Creswell and Plano Clark [11], [12].

A mixed-methods design is appropriate because the study looks at both measurable patterns and the reasons behind them. Quantitative ratings can show whether platforms with stronger AI transparency features tend to receive higher trust and adoption scores, but those numbers alone cannot explain why participants responded the way they did. Follow-up interviews can provide more detail about which features increased or reduced trust and why a participant might trust a platform but still not want to use it. The qualitative phase therefore helps explain the quantitative results and provides a more complete understanding of how transparency may influence trust and adoption.

The quantitative phase uses a repeated-measures, within-subject comparison because each participant would evaluate all five platforms. This allows each participant to serve as his or her own comparison across systems and reduces some variability caused by individual differences in technology confidence or LMS experience. Platform order would be counterbalanced to reduce learning, order, and fatigue effects.

B. Platforms, Materials, and Testing Environment

Five AI-powered LMS platforms would be evaluated: Coursebox AI, LearnWorlds, Mindsmith, TalentLMS, and Moodle with selected AI plugins. These systems were selected because they represent different approaches to integrating AI into online course creation, content generation, learning support, and LMS workflows.

A standardized sample course would be created in each platform. To the extent allowed by each system, the course would use the same title, learning objectives, instructional text, AI prompt, learning activity, sample learner records, and progress information. Fictional users and researcher-created content would be used instead of real institutional student data. Standardization is important because differences in course content could otherwise be mistaken for differences caused by the interface.

The researcher would document the platform version, subscription level, account type, AI features, and date of testing. For Moodle, the exact Moodle version, AI plugin name, plugin version, and enabled features would also be recorded. Participants would use the same browser, screen size, computer configuration, and internet connection whenever possible. These controls would not make the systems technically identical, but they would reduce avoidable environmental variation.

Before the main study, the tasks, transparency rubric, survey questions, and interview guide would be pilot tested with at least one student and one instructor who would not be included in the primary analytical sample. Pilot testing would be used to identify unclear instructions, tasks that cannot be completed equivalently across platforms, technical failures, and survey items that participants interpret inconsistently.

C. Participants

Approximately 10–15 university students and 3–5 university instructors would be recruited through convenience sampling. All participants would be at least 18 years old and would have prior experience with an LMS such as Canvas, Moodle, Blackboard, or a comparable platform. Student participants would be required to have completed or currently be enrolled in at least one LMS-supported course. Instructor participants would be required to have experience creating, organizing, or managing course content in an LMS.

Before testing, participants would complete a brief background questionnaire covering role, prior LMS experience, frequency of LMS use, familiarity with generative AI, and general frequency of AI-tool use. These variables would be used descriptively to characterize the sample and to help interpret unusual response patterns. The study is exploratory; the small

convenience sample would not be treated as representative of all students and instructors.

The qualitative phase would use purposive sampling from the original quantitative participants. Approximately six to eight participants would be selected after reviewing the quantitative results. Selection would intentionally include participants with high trust/high adoption, low trust/low adoption, and conflicting combinations such as high trust/low adoption or low trust/high adoption. Both students and instructors would be represented when possible. This connected sampling strategy is important because the interviews are intended to explain the quantitative results rather than create an unrelated qualitative dataset.

D. Independent Variable: AI-Transparency UX/UI Strength

The independent variable would be the strength of user-visible AI-transparency UX/UI features in each platform. Four components would be rated:

- 1) *AI labeling*: whether AI-generated or AI-modified content is clearly identified at the point where users encounter it.
- 2) *Explanations*: whether the interface provides understandable information about why an output, recommendation, feedback item, or action was produced.
- 3) *Source visibility*: whether users can see or access the sources, evidence, or supporting materials associated with AI-generated information when such sources are applicable.
- 4) *Uncertainty indicators*: whether the interface communicates confidence, uncertainty, limitations, or the need for verification rather than presenting all output with equal certainty.

Each component would be scored from 1 (very weak or absent) to 5 (strong and consistently implemented) using a standardized rubric with behavioral anchors. The four component scores would be averaged to create an overall AI-transparency score for each platform. Component scores would also be retained separately because two platforms could receive similar overall scores while using different transparency strategies.

To improve rating reliability, a second trained reviewer would independently score the same platform features using the same rubric. Reviewers would compare scores and document disagreements before resolving them through discussion. Keeping the original independent ratings as part of the audit record would make the scoring process more transparent and reduce the risk that the final platform classification reflects only one researcher's subjective impression.

E. Dependent Variables and Instruments

The primary dependent variables are user trust and adoption intention. Both constructs would be measured after participants use each platform.

Trust would be measured through a brief multi-item five-point Likert-type scale. Items would ask participants whether they consider AI-supported information reliable, whether they

feel confident in evaluating and using AI output from the platform, and whether the platform provides enough information for them to responsibly judge AI-supported results. Responses would range from strongly disagree to strongly agree and would be averaged to create a mean trust score for each participant-platform combination.

Adoption intention would be measured separately using multi-item five-point Likert-type statements about willingness to use the system in a future course, willingness to recommend its AI-supported features, and willingness to rely on the platform as part of normal learning or teaching work. Keeping adoption separate from trust allows the study to identify cases where a participant trusts a system but finds it too difficult to use, or likes its convenience but remains uncertain about its reliability.

The standard 10-item System Usability Scale would be administered after each platform as a secondary measure of perceived usability [13]. SUS scores would be converted to the established 0–100 scale. SUS is not part of the primary hypothesis; instead, it provides context for interpreting adoption. For example, a transparent platform could receive high trust but low adoption because of difficult navigation or poor workflow design.

An observation form would record task completion, requests for assistance, notable navigation problems, and spontaneous participant comments. These metrics would be secondary contextual information rather than primary dependent variables because the narrowed research focus is AI transparency, trust, and adoption.

F. Procedure

After informed consent, participants would complete the background questionnaire and receive a standardized orientation. The orientation would explain the study tasks and basic navigation expectations without teaching participants exactly how to complete the assigned actions.

Each participant would then evaluate all five platforms in a counterbalanced order. Student tasks would include locating course materials, identifying or accessing AI-generated learning content, completing a short learning activity, and viewing AI-supported feedback or progress information. Instructor tasks would include organizing course material, generating or editing AI-supported content, reviewing how the system communicates the AI's role, and viewing learner progress or analytics when available.

Immediately after using each platform, participants would complete the trust scale, adoption-intention scale, and SUS. Immediate measurement is intended to reduce recall error and make ratings reflect the platform just experienced. Participants could also provide a brief open comment after each system, but the primary qualitative phase would occur only after quantitative analysis.

After all participants finish the five-platform comparison, the researcher would calculate platform transparency scores and participant outcomes. Six to eight participants would then be selected for semi-structured interviews. Interview questions

would be tailored in part to the participant's quantitative pattern. For example, a participant who gave a platform high trust but low adoption would be asked what created confidence and what prevented willingness to use the system. A participant who gave unexpectedly low trust to a highly transparent platform would be asked which transparency cues were noticed and whether any felt confusing, excessive, or unconvincing.

G. Quantitative Data Analysis

Descriptive statistics would first summarize the transparency component scores, overall transparency scores, trust scores, adoption-intention scores, and SUS scores for each platform. Means, medians, standard deviations, and distributions would be inspected. Student and instructor results would also be summarized separately because the two groups perform different LMS tasks and may have different priorities.

Because the same participants would rate all five platforms, separate Friedman tests would be used to determine whether trust, adoption intention, and SUS scores differ significantly across platforms. If a Friedman test is significant, follow-up paired comparisons using Wilcoxon signed-rank tests would identify which platforms differ, with an adjustment for multiple comparisons.

Exploratory linear mixed effects models would be used because each participant would rate all five platforms. The overall AI transparency score would be used to determine whether platforms with stronger transparency features tend to receive higher trust and adoption ratings. The overall AI transparency score would be used to determine whether platforms with stronger transparency features tend to receive higher trust and adoption ratings. SUS scores would be included to examine whether general usability helps explain trust or adoption in addition to AI transparency. Because the instructor sample is expected to be small, differences between students and instructors would mainly be described separately rather than treated as a primary statistical comparison.

Statistical significance would be evaluated at $\alpha = .05$. A nonsignificant result would not necessarily mean that AI transparency is unrelated to trust or adoption. A nonsignificant result would not necessarily mean that AI transparency is unrelated to trust or adoption. It could also mean that a sample was too small, the platforms were too similar, or the measures were not sensitive enough to detect a clear relationship.

Most importantly, the models would be interpreted as tests of association rather than causation. Participants are not randomly assigned to different versions of the same interface, and transparency features are not independently manipulated. Platforms also differ in many ways beyond transparency. Therefore, a positive relationship between transparency scores and trust would support H_1 as an association but would not prove that a specific transparency feature caused the difference.

H. Qualitative Data Analysis

only with separate participant consent and then transcribed for analysis. The interview responses would be analyzed using

thematic coding. Initial codes would focus on topics related to the research questions, including AI labeling, explanations, source credibility, uncertainty indicators, perceived reliability, user confidence, verification, and willingness to adopt the platform. Additional themes would be added if participants raised important ideas that were not included in the original coding categories.

The analysis would compare themes among participants with high, low, or conflicting trust and adoption ratings. Both common patterns and differences between participants would be considered. For example, an uncertainty indicator might increase trust for one participant because it shows that the system is being honest about its limitations, while another participant might become less willing to rely on the system. These differences could help explain why the same transparency feature may affect users in different ways.

The researcher would keep a record of the coding process and connect each theme to de identified inter response and participant rating patterns. This would help keep the analysis organized and make it easier to explain how the final conclusions were developed from the data.

I. Mixed-Methods Integration

Integration would occur at the design, methods, and interpretation levels [11], [12]. At the design level, the qualitative phase is included specifically to explain the quantitative relationship between transparency, trust, and adoption. At the methods level, quantitative results determine which participants are interviewed and which findings receive follow-up questions. At the interpretation level, quantitative and qualitative results would be combined in a joint display and a side-by-side discussion.

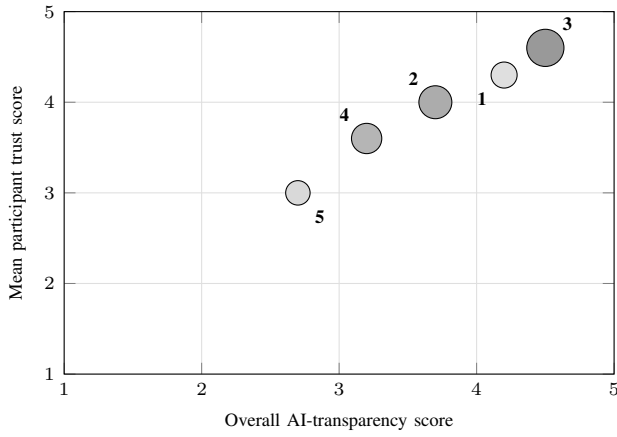
It would show the overall AI transparency score, the four individual transparency scores, average trust and adoption intention ratings, the average SUS score, and the main themes from participant interviews. These results would then be compared to see where the quantitative and qualitative findings support each other, add more detail, or show different patterns.

Figure 1 illustrates the planned integration of platform-level quantitative findings with qualitative explanations. Because data have not yet been collected, all values and qualitative statements shown in the figure are hypothetical and are included only to demonstrate the planned integration format.

As illustrated in Fig. 1, a possible meta-inference is that stronger AI transparency may support higher trust and adoption but that transparency does not operate independently of usability. For example, participants may value visible sources and explanations while still rejecting a platform with confusing navigation. Conversely, a highly usable interface may encourage adoption even when users express concern about opaque AI processes. Mixed-method integration is intended to distinguish these situations instead of reducing them to a single average score.

J. Validity, Limitations, and Ethical Considerations

Several steps would be used to strengthen the study. Standardized course content and tasks would reduce content-



Visual encoding: Larger bubbles indicate higher mean adoption intention. Darker bubbles indicate higher mean SUS scores.

#	Platform	Adoption	SUS	Illustrative QUAL Explanation
1	Coursebox AI	3.1	54	Clear AI features supported trust, but difficult navigation reduced willingness to adopt the platform.
2	LearnWorlds	4.1	76	Visible sources increased credibility and helped participants verify AI-generated information.
3	Mindsmith	4.5	82	Clear explanations made AI output easier to understand and evaluate.
4	TalentLMS	3.7	73	Participants found the interface usable but wanted more explanation of how AI results were produced.
5	Moodle with AI	2.9	58	Participants valued the system's flexibility but found its configuration confusing.

Fig. 1. Illustrative joint display of platform-level AI transparency, trust, adoption intention, usability, and qualitative explanations. The horizontal axis represents the researcher-rated overall AI-transparency score, and the vertical axis represents the mean participant trust score. Bubble size represents mean adoption intention, while bubble darkness represents the mean System Usability Scale score. All values and explanations are hypothetical.

related variation. Counterbalancing would reduce order and fatigue effects. Pilot testing would identify unclear procedures. A second reviewer would score transparency features to reduce single-rater bias. Trust, adoption, and usability would be measured as separate constructs. Qualitative sampling would intentionally include conflicting cases so that results that do not fit the main quantitative pattern are investigated rather than ignored.

The study nevertheless has important limitations. The small convenience sample limits generalizability, especially for instructors. Short testing sessions may not represent trust that develops over an entire semester. Platform versions, subscription tiers, and AI feature availability may change. The five platforms differ in functionality as well as transparency, creating potential confounding. The four transparency features may co-occur, making it difficult to isolate the contribution of any single component. Adoption intention is also a self-reported outcome rather than actual long-term use. Finally, with only five platform-level transparency profiles, mixed-effects models should be considered exploratory.

Before recruitment or data collection, the study would be

submitted to the university's Institutional Review Board for review or determination. Participation would be voluntary, and participants could withdraw without penalty. Each participant would receive a study identification number, and names would not be included in the analytical dataset. Consent records would be stored separately from research data.

Testing accounts would use fictional users and researcher-created materials. No participant passwords, grades, private course records, or personally identifiable student information would be collected. Electronic files would be stored in a password-protected, university-approved location. Any audio or screen recording would require separate consent.

The study also raises a professional responsibility beyond basic privacy protection. Transparency features should help users make informed judgments rather than create the appearance of certainty. AI labels should accurately identify AI involvement; source displays should not imply verification when a source does not support the generated claim; and confidence or uncertainty indicators should not overstate model reliability. Ethical interface design should therefore support calibrated trust—neither blind acceptance nor automatic rejection of AI—and preserve meaningful human judgment in educational decisions.

IV. EXPECTED RESULTS AND DISCUSSION

Based on the literature and the logic of the proposed model, platforms with stronger combinations of AI labeling, explanations, source visibility, and uncertainty communication are expected to receive higher average trust ratings. If users can see when AI is involved, understand the basis of an output, verify relevant sources, and recognize limitations, they should be better positioned to judge the system rather than treating it as an unexplained black box. This expectation is consistent with broader trust research identifying explainability and transparency as important antecedents of human trust in AI [4].

Adoption intention is also expected to be positively associated with transparency, but the relationship may be weaker than the relationship with trust. Adoption is likely to depend on both confidence and practical usability. A platform could communicate AI processes very clearly but still be rejected because routine tasks take too long or the interface is difficult to navigate. Conversely, a polished and convenient system may receive strong adoption ratings despite reservations about how its AI works. Including SUS allows the study to examine this distinction rather than assuming that transparency alone determines willingness to use a platform.

The qualitative interviews are expected to show that the four transparency features do not influence users in identical ways. Source visibility may be especially important when users want to verify factual or instructional claims. Explanations may help users understand why a recommendation was produced. AI labels may establish basic awareness and prevent users from mistaking generated content for fully human-authored material. Uncertainty indicators may have a more complex

effect: they could initially lower confidence in a particular answer while increasing longer-term trust in the system because the interface communicates limitations honestly. This would represent calibrated trust rather than simply “more” trust.

Differences may also appear between students and instructors. Instructors may place greater emphasis on verification, source quality, editing control, and responsibility for content presented to students. Students may place greater emphasis on clarity, convenience, and whether an explanation helps them decide when to rely on AI-generated material. Because the instructor sample is small, such differences would be treated as exploratory rather than definitive.

If the quantitative results show a significant positive association between transparency and both dependent variables, H_0 would be rejected in favor of H_1 within the limits of the study. A non-significant result would also be informative. It could indicate that transparency is less influential than expected, that users prioritize other design factors, or that the platforms do not differ enough in transparency to detect a relationship. Follow-up qualitative interviews would be particularly valuable in that situation because they could reveal whether participants noticed the transparency features at all or interpreted them differently than the researcher intended.

The study could contribute to research and practice by shifting attention from the existence of AI features to how those features are communicated. Tian and Choo demonstrate that generative AI can improve content quality and personalization [2]; the proposed study asks what must happen at the interface for users to understand and responsibly adopt such capabilities. The results could help LMS developers prioritize transparency patterns, help universities evaluate AI-enabled platforms during procurement, and help instructors choose tools that support informed rather than blind reliance on AI.

V. CONCLUSION AND FUTURE WORK

This study proposes an explanatory sequential mixed-methods evaluation of the relationship between AI-transparency UX/UI features, user trust, and adoption intention in AI-powered LMS platforms. It extends prior research on the technical and educational benefits of generative AI by focusing on the point where AI capability meets human judgment: the user interface. Five platforms would be compared using a standardized transparency rubric centered on AI labeling, explanations, source visibility, and uncertainty indicators. Student and instructor trust and adoption ratings would be analyzed quantitatively, with general usability treated as an important contextual factor. Follow-up interviews would then explain why specific transparency features strengthened, weakened, or failed to influence user confidence and willingness to adopt a platform.

The expected contribution is evidence that AI transparency should be treated as a measurable design characteristic rather than only an abstract ethical principle. Stronger transparency is expected to be associated with higher trust and adoption, but the relationship is likely to depend on overall usability and on how users interpret individual design features. The

study would therefore provide both quantitative comparisons and qualitative explanations that can inform more responsible LMS design and institutional technology decisions.

Two areas are especially important for future research. First, a controlled experiment could test AI labels, explanations, source visibility, and uncertainty indicators within the same LMS. Using one platform would make it easier to determine which transparency features have the strongest effect on trust and adoption without differences between platforms influencing the results. Second, a larger study across multiple universities could follow students and instructors over a full academic term. This would help show whether AI transparency is related not only to initial willingness to use a system, but also to actual use, continued adoption, verification behavior, and changes in trust over time.

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